

FINAL REPORT

Machine Learning Based Diabetes Risk Prediction System Using Clinical Data

Problem Id: 14

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Problem Description

Diabetes is a serious health condition in which the blood sugar level of a person becomes higher than normal. If diabetes is not detected early, it can lead to many health problems such as heart disease, kidney problems, vision issues, and other complications.

In this project, a Machine Learning Based Diabetes Risk Prediction System is developed using clinical data. The main aim of this system is to predict whether a person is diabetic or non-diabetic based on important health-related features.

The system uses patient details such as age, BMI, HbA1c level, and blood glucose level for prediction. These values are useful because they are closely related to diabetes risk.

The project follows a complete machine learning pipeline which includes data collection, data preprocessing, feature selection, model training, model evaluation, and final prediction.

Three machine learning classification algorithms are used:

- Logistic Regression**
- K-Nearest Neighbours**
- Support Vector Machine**

The final output of the system is:

**0 → Non-Diabetic
1 → Diabetic**

This system can be useful as a basic screening tool for diabetes risk prediction. It can help in giving an initial idea about diabetes risk, but it cannot replace proper medical testing or doctor consultation.

Dataset Description

The dataset used in this project is a diabetes prediction dataset. It contains clinical and lifestyle-related information of patients.

The dataset contains 100000 records and 9 columns. Each row represents one patient record. The dataset includes both numerical and categorical features.

The columns present in the dataset are:

Column Name	Description
gender	Gender of the patient
age	Age of the patient
hypertension	Shows whether the patient has hypertension or not
heart_disease	Shows whether the patient has heart disease or not
smoking_history	Smoking history of the patient
bmi	Body Mass Index of the patient
HbA1c_level	Average blood sugar level indicator
blood_glucose_level	Blood glucose level of the patient
Outcome	Target variable

The target variable is Outcome.

It contains two values:

0 → Non-Diabetic

1 → Diabetic

Out of 100000 records:

- 91500 patients are non-diabetic
- 8500 patients are diabetic

This shows that the dataset is imbalanced because the number of non-diabetic patients is much higher than diabetic patients.

Sample Dataset First Few Records

gender	age	hypertension	heart_disease	smoking_history	bmi	HbA1c_level	blood_glucose_level
Female	80.0	0	1	never	25.19	6.6	140
Female	54.0	0	0	No Info	27.32	6.6	80
Male	28.0	0	0	never	27.32	5.7	158
Female	36.0	0	0	current	23.45	5.0	155
Male	76.0	1	1	current	20.14	4.8	155

Dataset Analysis and Key Insights

Before training the models, the dataset was analysed to understand the data quality, structure, and important patterns.

Missing Values

The dataset was checked for missing values using Pandas. No missing values were found in the dataset. Therefore, no major missing value treatment was required.

Class Distribution

The dataset contains more non-diabetic records than diabetic records. There are 91500 non-diabetic records and 8500 diabetic records.

This means the dataset is imbalanced. Due to this, only accuracy is not enough for model evaluation. Precision, recall, and F1-score are also important.

Feature Importance

The most useful features selected for final model training are:

- **HbA1c_level**
- **blood_glucose_level**
- **bmi**
- **age**

HbA1c level and blood glucose level are directly related to sugar level in the body. BMI and age also help in understanding diabetes risk.

Feature Distribution

The important numerical features have different value ranges. For example, age is measured in years, BMI is a body mass index value, HbA1c level is measured as a sugar-level indicator, and blood glucose level is measured separately.

Because these features have different scales, feature scaling was applied before training the model.

Key Observation

Higher HbA1c level and blood glucose level are strongly related to diabetes risk. BMI and age also support the prediction because diabetes risk can increase with unhealthy body weight and increasing age.

Steps of Implementation

The project follows a structured machine learning pipeline.

1. Data Collection

The dataset was collected in CSV format and loaded into Jupyter Notebook using Pandas.

The dataset contains patient details such as gender, age, hypertension, heart disease, smoking history, BMI, HbA1c level,

blood glucose level, and Outcome.

After loading the dataset, the shape of the dataset and first few rows were displayed to understand the structure of the data.

The dataset shape was:

100000 rows and 9 columns

2. Data Preprocessing

Data preprocessing was performed to prepare the dataset for machine learning model training.

The following steps were performed during preprocessing:

- Missing values were checked**
- Outliers were removed using the IQR method**
- Categorical columns were encoded**
- Final selected features were separated**
- Feature scaling was applied using Standard Scaler**

Outliers were removed from important numerical columns such as age, BMI, HbA1c level, and blood glucose level.

Before outlier removal, the dataset had:

100000 rows

After outlier removal, the dataset had:

90387 rows

Total outliers removed:

9613 rows

Feature scaling was applied because models like KNN and SVM are distance-based models, and they work better when all features are on a similar scale.

3. Feature Selection

Feature selection was performed to reduce unnecessary columns and use only the most useful features for prediction.

The final selected features are:

Selected Feature	Reason
HbA1c_level	Important indicator of long-term blood sugar level
blood_glucose_level	Directly related to diabetes
bmi	Helps identify weight-related diabetes risk
age	Diabetes risk may increase with age

Only these four features were used for final model training and prediction.

This makes the model simple, faster, and easier to understand.

4. Model Selection and Training

Three machine learning classification models were selected and trained.

Logistic Regression:

Logistic Regression is a simple and commonly used classification algorithm. It was used as a baseline model in this project.

K-Nearest Neighbours:

KNN classifies a new record by comparing it with nearby similar records. It works based on distance between data points.

Support Vector Machine:

SVM tries to find the best decision boundary between different classes. In this project, the RBF kernel was used.

The dataset was split into training and testing data.

- Training Data: 80%
- Testing Data: 20%

Stratified splitting was used so that the class distribution `

5. Model Evaluation

The models were evaluated using different performance metrics.

Accuracy:

Accuracy measures how many predictions are correct out of total predictions.

Accuracy = Correct Predictions / Total Predictions

Precision:

Precision measures how many predicted diabetic cases are actually diabetic.

Precision = TP / (TP + FP)

Recall:

Recall measures how many actual diabetic cases are correctly predicted by the model.

Recall = TP / (TP + FN)

F1-Score:

F1-score is the balance between precision and recall. It is useful when the dataset is imbalanced.

F1-Score = $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

Confusion Matrix:

A confusion matrix shows actual and predicted values. It helps in understanding correct and incorrect predictions.

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

Where:

TP = True Positive

TN = True Negative

FP = False Positive

FN = False Negative

6. Model Results

The performance of all three models was compared using accuracy, precision, recall, and F1-score.

Model	Accuracy	Precision	Recall	F1-Score
1. Logistic Regression	0.8329	0.2121	0.8687	0.3409
2. KNN	0.9684	0.8475	0.4449	0.5835
3. SVM	0.8290	0.2146	0.9166	0.3478

From the results, KNN achieved the highest accuracy and F1-score.

7. Best Model Selection

**The best model selected for this project is:
K-Nearest Neighbours**

KNN was selected because it gave the best overall performance among the three models.

KNN performance:

- Accuracy: 0.9684
- Precision: 0.8475
- Recall: 0.4449
- F1-Score: 0.5835

Although SVM achieved higher recall, its precision was low. Logistic Regression also gave good recall but low precision.

KNN gave the best balance between accuracy, precision, and F1-score. Therefore, KNN was selected as the final model for prediction.

8. Prediction System

After selecting the best model, a simple prediction system was created.

The prediction system takes four input values from the user:

- HbA1c_level
- blood_glucose_level
- bmi
- age

These input values are converted into a DataFrame and then scaled using the same StandardScaler used during training.

After scaling, the trained KNN model predicts whether the patient is diabetic or non-diabetic.

The final prediction output is:

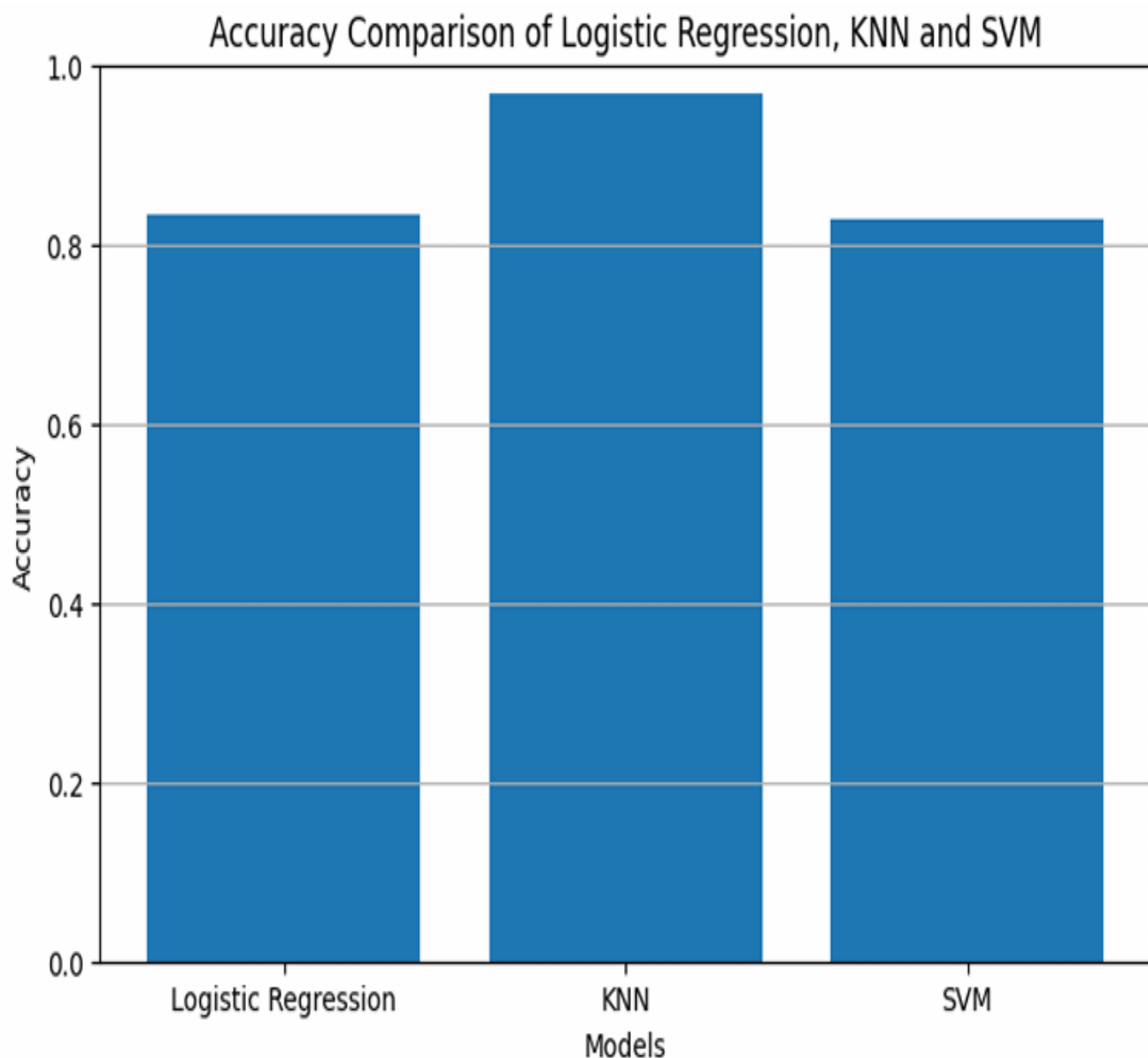
0 → Patient is predicted as Non-Diabetic

1 → Patient is predicted as Diabetic

Example output:

Prediction from our best Model (KNN): 0

The patient is predicted as Non-Diabetic



Expected Modules / Libraries Used

Inbuilt Libraries:

NumPy

NumPy was used for numerical operations and array handling.

Pandas

Pandas was used for loading the dataset, checking missing values, handling data, and creating DataFrames.

Matplotlib

Matplotlib was used for plotting graphs such as accuracy comparison graph and confusion matrix display.

Scikit-learn

Scikit-learn was used for train-test split, feature scaling, model training, and model evaluation.

Custom Modules:

Data Preprocessing Module

This part handles missing value checking, outlier removal, encoding, and scaling.

Feature Selection Module

This part selects the important features used for final model training.

Model Training Module

This part trains Logistic Regression, KNN, and SVM models.

Model Evaluation Module

This part evaluates all models using accuracy, precision, recall, F1-score, classification report, and confusion matrix.

Prediction Module

This part takes user input and predicts whether the patient is diabetic or non-diabetic.

Expected Evaluation Metrics and Plots

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score
- Classification Report
- Confusion Matrix

These metrics were used to compare the performance of Logistic Regression, KNN, and SVM.

Visualization Plots

Confusion Matrix

Confusion matrix was used to show correct and incorrect predictions for each model.

Accuracy Comparison Graph

Accuracy comparison graph was used to compare the accuracy of Logistic Regression, KNN, and SVM.

Required Topics from the Subject

The following subject topics were used in this project:

- Arrays and Data Structures
- Functions
- File Handling
- Basic Statistics
- Data Preprocessing
- Feature Scaling
- Classification Algorithms
- Model Evaluation

NumPy arrays and Pandas DataFrames were used for storing and processing data. Functions were used to organize model evaluation. File handling was used to load the CSV dataset. Basic statistics such as quartiles and IQR were used for outlier removal.

Platform Required

- **Python**
- **Jupyter Notebook**
- **Visual Studio Code**

Books and Link Sources

- 1. Kaggle Diabetes Prediction Dataset**
- 2. Scikit-learn Documentation**
- 3. Pandas Documentation**
- 4. GeeksforGeeks Machine Learning Tutorials**

Final Output

The final output of the system is:

- **0 → Non-Diabetic**
- **1 → Diabetic**

The final trained KNN model predicts diabetes risk using selected clinical input values: HbA1c level, blood glucose level, BMI, and age.

Conclusion

In this project, a diabetes risk prediction system was developed using machine learning. The dataset was first loaded and analysed. After that, preprocessing steps like missing value checking, outlier removal, categorical encoding, feature selection, and feature scaling were performed.

Three classification models were trained: Logistic Regression, KNN, and SVM. After training and evaluation, KNN gave the best performance based on accuracy and F1-score.

Therefore, KNN was selected as the final model. The final system takes clinical input values and predicts whether a person is diabetic or non-diabetic.

This project shows how machine learning can be used in healthcare-related prediction tasks. It can be used as a basic support system for early diabetes risk screening.